

James Friedenber

Senior Software Engineer

Albany, New York | jamesfriedenber@gmail.com | (845) 245-9249

[LinkedIn](#) | [Portfolio](#) | [Github](#)

High-impact Software Engineer with 7+ years of total experience, with 4+ years at Microsoft building scalable live-event systems, cloud infrastructure, and developer tooling. Proven track record of driving **8x server capacity increases**, **cutting Azure infrastructure costs by 88%**, and delivering automated pipelines that save hundreds of engineering labor hours per release cycle. Well-versed in full-stack TypeScript/React ecosystems and high-scale Azure cloud environments.

PROFESSIONAL EXPERIENCE

Software Engineer, Microsoft - Remote, USA

March 2022 - June 2026

- Engineered Minecraft: Bedrock Edition's first automated multiplayer performance testing framework, achieving an 8x increase in server capacity (10 to 80 players) and cutting Azure infrastructure costs by 88% (~\$20k+ saved per live event).
- Developed the foundational prototype and gameplay systems for the Minecraft Legends promotional live event, reaching 3M+ players and driving an estimated \$6M in marketing value for the franchise launch.
- Modernized the MCTools codebase and expanded creator content validation suites, replacing manual workflows with automated pipelines to cut hundreds of engineering labor hours per release cycle.
- Designed and implemented a telemetry and observability framework for MCTools using Azure Application Insights, enabling custom event instrumentation, real-time monitoring, issue detection, and operational visibility across the platform.
- Developed a localization framework, enabling MCTools to be available in 20+ languages and bringing it into compliance with the minecraft.net domain.
- Mentored and onboarded new engineers to the Minecraft web and platform technology stack, providing guidance on React, TypeScript, deployment workflows, telemetry tooling, and development practices.

Software Design Engineer in Test, DIGARC - Remote, USA

March 2021 - March 2022

- Created detailed test plans and tracked quality initiatives using Jira for an Agile team of developers, testers, and a PM.
- Engineered and maintained several hundred automated E2E tests, ensuring reliable frontend regression coverage.
- Spear-headed the implementation of automation testing frameworks Vitest and Cypress

Software Design Engineer in Test, Andromeda Systems Inc. - Jacksonville, Florida

August 2019 - March 2021

- Developed over 500 frontend automated tests for an inventory management application used by the U.S. Navy for aircraft maintenance.
- Designed and built dozens of Angular components using TypeScript in an Agile environment.

PERSONAL PROJECTS

jeemzu.me - Full-Stack Portfolio & Browser Games Platform

- Architected and shipped a full-stack web platform (React 19, ASP.NET Core 8, PostgreSQL) with Docker multi-stage builds, GitHub Actions CI/CD with cross-repo type generation, Azure Container Registry deployment, and Netlify SPA hosting with security headers.
- Built 5 browser games using Phaser 3 and 2 native-performance games in C++20 compiled to WebAssembly via Emscripten, with a shared modal system supporting pause, difficulty, volume, and game-over stat overlays.
- Implemented JWT authentication with httpOnly cookie refresh-token rotation, role-based access control, and a score upsert leaderboard system with per-user unique constraints and composite indexing.
- Designed a custom platformer level editor with localStorage persistence, a bundled level manifest, and a C++ level API bridge that translates JSON level schemas into WASM function calls at runtime.

JeemzuAI - Multi-Agent RAG Chat System

- Engineered a Retrieval-Augmented Generation pipeline using Semantic Kernel, GPT-4o-mini, and text-embedding-3-small with pgvector HNSW cosine indexing on 1536-dimensional embeddings for sub-second semantic search over a personal knowledge base.
- Built a Python multi-agent orchestration layer using LangGraph and FastAPI with 5 specialized agents (router, knowledge, game stats, web search, synthesizer) that execute in parallel and synthesize responses from multiple data sources.
- Implemented tool-calling agents that query live .NET API endpoints for real-time leaderboard data and user profiles, plus Tavily web search integration for general knowledge questions.

SKILLS

Languages: TypeScript/JavaScript, C#, C++, Python

Frontend: React 19, Phaser 3, MUI, Zustand, Zod, Wouter, Emscripten/WebAssembly

Backend: ASP.NET Core 8, .NET, Node.js, REST API Development, Entity Framework Core

Data & AI: PostgreSQL, pgvector, RAG Pipelines, OpenAI API, Semantic Kernel, LangGraph, LangChain, Vector Embeddings, HNSW Indexing

Cloud & DevOps: Azure App Service, Azure PostgreSQL, Azure Application Insights, Docker, Docker Compose, GitHub Actions, CI/CD, Netlify

Testing & Quality: Playwright, Selenium, Vitest, React Testing Library, Jest, Postman

EDUCATION

Rochester Institute of Technology - BS in Game Design and Development

August 2015 - May 2019